MANO HARSHA DAVULURI

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PROFESSIONAL SUMMARY

Results-driven **NITian** and **AI & Data Engineer** with 3+ years of experience building production Generative AI applications and scalable data platforms. Specializes in agentic AI (MCP, LLM tool-calling, RAG with vector search), LLMOps with MLflow, and FastAPI services on Databricks, with a strong foundation in Spark, AWS, data lakes, and ETL pipelines. Architected and launched an enterprise conversational AI assistant that enables natural-language data discovery across data assets, products, and domains. Proven track record of leading cross-functional teams and driving AI-led innovation with data governance at enterprise scale.

TECHNICAL SKILLS

Generative AI & Agents: Agentic AI, Multi-agent Systems (Agent-to-Agent), Retrieval-Augmented Generation (RAG), MCP (FastMCP), LangChain, LangGraph, OpenAI API, LLMs (Anthropic Claude, Meta Llama), Prompt Engineering, Guardrails, Embeddings, Vector Database (Databricks Vector Search), Hybrid Search (pgvector HNSW, RRF), Machine Learning.
LLMOps & AI Platforms: MLflow 3 (Tracing, LLM Judges), Model Serving, AI Gateway (Routing, Fallbacks), Databricks Apps, AI Tools (Claude Code, Cursor).
Cloud, Platforms & Databases: Databricks (Unity Catalog, Lakebase, Delta Lake, SQL Warehouses), AWS (S3, Lambda, EKS/Kubernetes, CloudWatch), PostgreSQL.
Data Engineering: Apache Spark, PySpark, Spark SQL, Delta Lake, Brickflow, SparkExpectations, DBT, Informatica CDGC.
Data Modeling: Dimensional Modeling, Star/Snowflake Schema, SCD Type 1/2, Medallion, Lakehouse, Data Governance.
Programming & DevOps: Python, SQL, FastAPI Microservices, Pydantic, Git, Docker, Jenkins CI/CD, OAuth 2.0 (Okta, M2M).

PROFESSIONAL EXPERIENCE

Nike | *Software Engineer II – AI* *Bengaluru, India | Jan 2026 – Present*

Compass AI – Agentic chatbot for Nike's enterprise data catalog

- Developed an **MCP (Model Context Protocol)** server on Databricks Apps for a **LangChain AI agent harness** powered by **Claude Sonnet 4.5 (Anthropic)**, growing it to **12 LLM-callable tools**, incl. a **multi-agent** (agent-to-agent) tool calling Nike's AirTalk chatbot on the user's behalf via **Okta OAuth**, and tools for AI-proposed data products, docs lookup, and catalog stats.
- Created retrieval tools over **Lakebase (Postgres)** with **hybrid semantic search** (pgvector + full-text via **Reciprocal Rank Fusion**) and safe SQL access; replaced an external REST API with direct table queries, removing a network hop and its auth.
- Instrumented end-to-end **observability** with **MLflow 3 tracing**, propagating request/session IDs from FastAPI to the MCP server via HTTP headers to group each multi-turn chat into one session, scored by **LLM-as-a-judge** evaluators (completeness, role adherence, tool-call efficiency, user frustration) with time-to-first-token monitoring.
- Configured **Databricks AI Gateway** with weighted **traffic splitting** (70% Claude Sonnet 4.6, 30% Claude Opus 5.5) and session-sticky routing, plus an automatic **fallback** to Llama 3.3 70B for high availability; added Anthropic **prompt caching** to cut token cost and latency.
- Crafted the agent's **system prompt and guardrails**: scope restriction to Nike metadata, non-disclosure of internals, correct bot identity, and accurate deep links to products, assets, and metrics.
- Productionized the MCP server across **dev/QA/prod** with **Jenkins CI/CD**, per-env config, **service-principal OAuth M2M**, and Cerberus secrets; retired a legacy MCP server and scheduled access-scope and vector index refresh jobs.
- Shipped **FastAPI** chatbot features: chat-history APIs (sessions, timestamped recent queries, per-session limits), a live health endpoint for UI "Online" status, and resilient ServiceNow ticketing, at a **90% test coverage** gate.

Epsilon (Publicis Groupe) | *Software Engineer I – Data & ML* *Bengaluru, India | Aug 2023 – Dec 2025*

- Designed a comprehensive Data Quality framework for the Prospect project, featuring automated discrepancy detection and real-time alerting mechanisms to ensure data integrity and compliance.
- Constructed scalable ETL/ELT data solutions processing over 150+ billion records daily into data lakes and warehouses using Spark, and trained ML models that enhanced decision-making, driving a 25% boost in quarterly revenue.
- Optimized and tuned Spark jobs for large-scale datasets (30+ TB), applying performance best practices and reducing infrastructure costs by 46%.
- Partnered with cross-functional teams to define data governance standards and set up CI/CD pipelines for data workflows.
- Owned end-to-end data pipelines spanning ingestion, extract-transform-load processes, data archiving, warehousing and recurring reporting using Apache Spark on Databricks Delta tables.

CERTIFICATIONS

- Databricks Certified Generative AI Engineer Associate – Databricks (Issued 2026)
- Databricks Certified Data Engineer Associate – Databricks (Issued 2024)

EDUCATION

National Institute of Technology (NIT) Karnataka | *Aug 2019 - May 2023*
Bachelor of Technology | Surathkal, Karnataka